Then it follows directly from $\frac{C_{t+1}^i}{C_t^i} = \exp\left[\eta_{t+1}^i y_{t+1} - \frac{y_{t+1}^2}{2}\right]\frac{C_{t+1}}{C_t}$ that

$$E_t\left[R_{j,t+1}\hat{\beta}\left(\frac{C_{t+1}}{C_t}\right)^{-\hat{\gamma}}\right] = 1, \tag{15.10.9}$$

where

$$\log\hat{\beta} = \log\beta + \frac{\gamma(\gamma+1)}{2}a \tag{15.10.10a}$$

$$\hat{\gamma} = \gamma - \frac{\gamma(\gamma+1)}{2}b. \tag{15.10.10b}$$

Formula (15.10.10b) implies that if $b < 0$, so that the cross-section dispersion of consumption growth increases during downturns in $\left(\frac{C_{t+1}}{C_t}\right)$, then $\hat{\gamma} > \gamma$. So in this case, estimates of γ constructed in usual ways from aggregate consumption data are distorted upwards.

Storesletten, Telmer, and Yaron (1998, 2004) and Cogley (1999) pursued some of the ideas of Constantinides and Duffie by using evidence from the panel study of income dynamics (PSID) to estimate the persistence of endowment shocks and the volatility of consumption innovations.

15.11. Affine risk prices

This section describes another response to the empirical problems encountered by the theory $E(m_{t+1}R_{j,t+1}) = 1$ under the stochastic discount factor (15.6.8)-(15.6.9) induced by time separable CRRA preferences, which we repeat here for convenience:

$$m_{t+1} = \exp\left(-r_t - \frac{1}{2}\sigma_c^2\gamma^2 - \gamma\sigma_c\varepsilon_{t+1}\right)$$

$$r_t = \rho + \gamma\mu - \frac{1}{2}\sigma_c^2\gamma^2.$$

This model of the stochastic discount factor asserts that exposure to the random part of consumption growth, $\sigma_c\varepsilon_{t+1}$, is the only possible source discrepancies among assets' expected rates of returns: ε_{t+1} is the only priced risk.

Another approach maintains $E(m_{t+1}R_{t+1}) = 1$ but divorces the stochastic discount factor m_{t+1} from consumption risk. Instead, this approach specifies a stochastic discount factor that (a) is analytically tractable (a test passed by

(15.6.4) or (15.6.8)-(15.6.9)), and (b) can be calibrated to fit observed asset prices without provoking skeptical comments about implausible parameter values and magnitudes of risk prices (a test critics like Lucas say (15.6.4) or Tallarini's (15.7.21)-(15.7.22) fails to pass).

The alternative approach imposes the law of one price via $E(m_{t+1}R_{t+1}) = 1$ (and often also the no-arbitrage principle via $m_{t+1} > 0$) but abandons a link between the stochastic discount factor and a consumption growth process. Instead, it posits a stochastic process for the stochastic discount factor that is not tightly linked to a theory about consumers' preferences, and then uses overidentifying restrictions from $E(m_{t+1}R_{j,t+1}) = 1$ for a set of N returns $R_{j,t+1}, i = 1, \ldots, N$, to let the data reveal risks and their prices.

The model has two components. The first is a vector autoregression that describes underlying risks ε_{t+1} and the evolution of the yield r_t on a one period risk free claim:[43]

$$z_{t+1} = \mu + \phi z_t + C\varepsilon_{t+1} \tag{15.11.1}$$

$$r_t = \delta_0 + \delta_1' z_t, \tag{15.11.2}$$

where ϕ is a stable $m \times m$ matrix, C is an $m \times m$ matrix, $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$ is an i.i.d. $m \times 1$ random vector, and z_t is an $m \times 1$ state vector. The second component is a vector of risk prices λ_t and an associated stochastic discount factor m_{t+1} defined via

$$\lambda_t = \lambda_0 + \lambda_z z_t \tag{15.11.3}$$

$$\log(m_{t+1}) = -r_t - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\varepsilon_{t+1}. \tag{15.11.4}$$

Here λ_0 is $m \times 1$ and λ_z is $m \times m$. The entries of λ_t that multiply corresponding entries of the risks ε_{t+1} are called risk prices, for reasons that we explain in the next subsection. The stochastic discount factor m_{t+1} is exponential quadratic in the state z_t as a result of the risk prices λ_t being affine in z_t. Evidently,

$$E_t(m_{t+1}) = \exp(-[\delta_0 + \delta_1' z_t]) = \exp(-r_t) \tag{15.11.5}$$

$$\text{std}_t(m_{t+1}) = E_t(m_{t+1})(\exp(\lambda_t'\lambda_t) - 1)^{\frac{1}{2}} \approx |\lambda_t|. \tag{15.11.6}$$

Equation (15.11.5) confirms that r_t is the yield on a risk-free one-period bond. That is why it is often called 'the short rate' in the literature on exponential

[43] Note that we are recycling notation by redefining ε_{t+1} here.

quadratic models of the stochastic discount factor. The free parameters of the vector autoregression for the $\{z_t\}$ process are (μ, ϕ, C) and the free parameters $(\delta_0, \delta_1, \lambda_0, \lambda_z)$ pin down the stochastic discount factor as a function of (z_t, ε_{t+1}).

15.11.1. An application

We gather insights by using the stochastic discount factor (15.11.4) to restrict a risky one-period gross return $R_{j,t+1}$. Our standard pricing formula is

$$E_t\left(m_{t+1}R_{j,t+1}\right) = 1. \tag{15.11.7}$$

Let $R_{j,t+1}$ be described by

$$R_{j,t+1} = \exp\left(\nu_t\left(j\right) - \frac{1}{2}\alpha_t\left(j\right)'\alpha_t\left(j\right) + \alpha_t\left(j\right)'\varepsilon_{t+1}\right) \tag{15.11.8}$$

or

$$\log R_{j,t+1} \sim \mathcal{N}\left(\nu_t\left(j\right) - \frac{1}{2}\alpha_t\left(j\right)'\alpha_t\left(j\right), \alpha_t\left(j\right)'\alpha_t\left(j\right)\right),$$

where $\nu_t(j)$ is a function of z_t that makes (15.11.7) become satisfied and

$$\alpha_t\left(j\right) = \alpha_0\left(j\right) + \alpha_z\left(j\right)z_t, \tag{15.11.9}$$

where $\alpha_0(j)$ is an $m \times 1$ vector and $\alpha_z(j)$ is an $m \times m$ matrix. In (15.11.8), the components of the loading vector $\alpha_t(j)$ express *exposures* of $\log R_{j,t+1}$ to corresponding components of the vector of risks ε_{t+1}. Equation (15.11.8) implies

$$E_t R_{j,t+1} = \exp\left(\nu_t\left(j\right)\right).$$

We want to know how λ_t and $\alpha_t(j)$ affect $\nu_t(j)$.

The formula for the mean of a log normal random variable implies that (15.11.7) becomes $\exp\big(\nu_t(j) - r_t - \alpha_t(j)'\lambda_t\big) = 1$ or

$$\nu_t\left(j\right) = r_t + \alpha_t\left(j\right)'\lambda_t. \tag{15.11.10}$$

According to (15.11.10), the net expected return $\nu_t(j)$ equals the net risk-free return r_t plus the transposed vector λ_t' of risk prices times the vector $\alpha_t(j)$ of the risk exposures of the logarithm of the gross return $R_{j,t+1}$. This outcome expresses how λ_t is a vector of risk prices that tell how the expected return on an asset depends on its exposures to the risks ε_{t+1}.

15.11.2. Affine term structure of yields

A good example of an exponential quadratic model of a stochastic discount factors is the model of the term structure of yields on risk-free claims constructed and estimated by Ang and Piazzesi (2003). Let $p_t(n)$ be the price at time t of a risk-free claim to one unit of the consumption good at time $t+n$. The one-period gross *return* on holding an $n+1$ period pure discount bond from t to $t+1$ is $R_{t+1} = \frac{p_{t+1}(n)}{p_t(n+1)}$. Evidently, $E_t\left(m_{t+1}R_{t+1}\right) = 1$ implies[44]

$$p_t\left(n+1\right) = E_t\left(m_{t+1}p_{t+1}\left(n\right)\right) \tag{15.11.11}$$

and

$$p_t\left(1\right) = E_t\left(m_{t+1}\right) = \exp\left(-\delta_0 - \delta_1' z_t\right) = \exp\left(-r_t\right). \tag{15.11.12}$$

From (15.11.11) and (15.11.12), it follows that

$$p_t\left(n\right) = \exp\left(\bar{A}_n + \bar{B}_n z_t\right), \tag{15.11.13}$$

where $(\bar{A}_n, \bar{B}_n)$ solve the system of difference equations

$$\bar{A}_{n+1} = \bar{A}_n + \bar{B}_n'\left(\mu - C\lambda_0\right) + \frac{1}{2}\bar{B}_n' CC'\bar{B}_n - \delta_0 \tag{15.11.14}$$

$$\bar{B}_{n+1}' = \bar{B}_n'\left(\phi - C\lambda_z\right) - \delta_1', \tag{15.11.15}$$

subject to the initial conditions $\bar{A}_1 = -\delta_0, \bar{B}_1 = -\delta_1$.

The *yield* to maturity on an n period pure discount bond is defined as

$$y_t\left(n\right) = -\frac{\log\left(p_t\left(n\right)\right)}{n}, \tag{15.11.16}$$

which is equivalent with $p_t(n) = \exp\left(-ny_t(n)\right)$. Evidently,

$$y_t\left(n\right) = A_n + B_n' z_t, \tag{15.11.17}$$

where $A_n = -\bar{A}_n/n, B_n = -\bar{B}_n/n$. Yields are affine functions functions of z_t.

This model for yields fits within a class of linear Gaussian models that can be estimated by maximum likelihood using methods described in section

[44] See exercise 15.14.

2.8 of chapter 2. See Ang and Piazzesi (2003) and Piazzesi (2005) for applications where some of the factors are interpreted in terms of a monetary policy authority's rule for setting a short rate.[45]

15.12. Risk-neutral probabilities

We return to the section 15.11 vector autoregression and short-rate equation (15.11.1)-(15.11.2), which for convenience we repeat here

$$\begin{aligned} z_{t+1} &= \mu + \phi z_t + C\varepsilon_{t+1} \\ r_t &= \delta_0 + \delta_1' z_t, \end{aligned}$$

where $\varepsilon \sim \mathcal{N}(0, I)$. We suppose that this structure describes the data generating mechanism. Finance economists call this the "physical measure" to distinguish it from what they call a "risk neutral measure", an object that we now describe. Evidently, under the physical measure generated by (15.11.1)-(15.11.2), the distribution of z_{t+1} conditional on z_t is $\mathcal{N}(\mu + \phi z_t, CC')$. As in section 15.11, define the vector of risk prices $\lambda_t = \lambda_0 + \lambda_z z_t$ and then define the following nonnegative random variable:

$$\frac{\xi_{t+1}^Q}{\xi_t^Q} = \exp\left(-\frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\varepsilon_{t+1}\right). \tag{15.12.1}$$

Evidently, $\frac{\xi_{t+1}^Q}{\xi_t^Q}$ is a log normal random variable with mean unity. Therefore, $\frac{\xi_{t+1}^Q}{\xi_t^Q}$ is a likelihood ratio can be used to twist the distribution of z_{t+1} conditional on z_t. Multiplying the conditional distribution of z_{t+1} by this likelihood ratio transforms it into the so-called "risk neutral" conditional distribution $z_{t+1} \sim \mathcal{N}(\mu + \phi z_t - C\lambda_t, CC')$ or

$$z_{t+1} \sim \mathcal{N}\left(\mu - C\lambda_0 + (\phi - C\lambda_z)\, z_t, CC'\right). \tag{15.12.2}$$

The risk neutral conditional distribution's twisting of the conditional mean of the original or "physical" measure from $\mu + \phi z_t$ to $\mu - C\lambda_0 + (\phi - C\lambda_z)z_t$ encodes

[45] Appendix C of this chapter uses methods of chapter 2 to bring out some of the implications of a simple affine term structure model of Backus and Zin (1994). Also see Chen and Scott (1993), Dai and Singleton (2000), and Piazzesi and Schneider (2006).

how the formula $E_t m_{t+1} R_{j,t+1} = 1$ adjusts expected returns for exposures to the vector of risks ε_{t+1}.

15.12.1. Asset pricing in a nutshell

Let E^P denote an expectation under the physical measure that nature uses to generate the data. Our key asset pricing equation is $E_t^P m_{t+1} R_{j,t+1} = 1$ for all returns $R_{j,t+1}$. Using (15.12.1), it is convenient to express the exponential quadratic stochastic discount factor (15.11.4) as

$$m_{t+1} = \frac{\xi_{t+1}^Q}{\xi_t^Q} \exp\left(-r_t\right),$$

where remember that r_t is the risk-free net short rate. Then the condition $E_t^P m_{t+1} R_{j,t+1} = 1$ is equivalent with $E_t^P \exp(-r_t) \frac{\xi_{t+1}^Q}{\xi_t^Q} R_{j,t+1} = 1$ or

$$E_t^Q R_{j,t+1} = \exp\left(r_t\right),$$

where E_t^Q is the conditional expectation under the risk neutral measure (15.12.2). Under the risk neutral measure, expected returns on all assets equal the risk-free return.

15.13. Distorted beliefs

Piazzesi, Salomao, and Schneider (2015) assemble survey evidence that suggests that economic experts' forecasts are systematically biased. Let $\{z_t\}_{t=1}^T$ be a record of observations on the state z and let $\{\check{z}_{t+1}\}_{t=1}^T$ be a record of one-period ahead expert forecasts. Let $\check{\mu}, \check{\phi}$ be regression coefficients in the least squares regression

$$\check{z}_{t+1} = \check{\mu} + \check{\phi} z_t + e_{t+1}, \tag{15.13.1}$$

where the residual e_{t+1} has mean zero, is orthogonal to z_t, and we assume that $E e_{t+1} e_{t+1}' = CC'$. By comparing estimates of the regression coefficients μ, ϕ in equation (15.11.1) that nature uses to generate the data with estimates of $\check{\mu}, \check{\phi}$ in (15.13.1) that describe the subjective beliefs of the experts, Piazzesi,

Salomao, and Schneider deduce that experts' beliefs are systematically distorted.

To organize the evidence about how the experts' subjective beliefs are systematically distorted relative to the physical measure, Piazzesi, Salomao, and Schneider let $\kappa_t = \kappa_0 + \kappa_z z_t$ and then define the likelihood ratio[46]

$$\frac{\xi_{t+1}^S}{\xi_t^S} = \exp\left(-\frac{1}{2}\kappa_t'\kappa_t - \kappa_t'\varepsilon_{t+1}\right). \tag{15.13.2}$$

Multiplying the conditional distribution of z_{t+1} under the physical measure by the likelihood ratio $\frac{\xi_{t+1}^S}{\xi_t^S}$ transforms it to the experts' subjective conditional distribution $z_{t+1} \sim \mathcal{N}(\mu + \phi z_t - C\kappa_t, CC')$ or

$$z_{t+1} \sim \mathcal{N}\left(\mu - C\kappa_0 + (\phi - C\kappa_z)\, z_t, CC'\right).$$

The discrepancy between the conditional mean of z_{t+1} under the physical measure $\mu + \phi z_t$, and the conditional mean $\mu - C\kappa_0 + (\phi - C\kappa_z)z_t$ under the subjective measure characterizes how the subjective S measure differs from the physical P measure that generates the data. In their regression (15.13.1) of experts' forecasts $\check{z}_{t+1}$ on z_t, Piazzesi, Salomao, and Schneider interpret their estimate of $\check{\mu}$ as an estimate of $\mu - C\kappa_0$ and $\check{\phi}$ as an estimate of $(\phi - C\kappa_z)$. In particular, they find that the experts' forecasts are formed as if the level and slope of the yield curve are more persistent than under the physical measure.

Piazzesi, Salomao, and Schneider explore the hypothesis that a representative agent with these distorted beliefs prices assets and makes returns satisfy

$$E_t^S m_{t+1}^* R_{j,t+1} = 1, \tag{15.13.3}$$

where E_t^S is a conditional expectation with respect to the subjective S measure rather than the physical measure and m_{t+1}^* is the stochastic discount factor of a representative agent having these subjective beliefs. In particular, Piazzesi, Salomao, and Schneider's representative agent with distorted beliefs has a stochastic discount factor

$$m_{t+1}^* = \exp\left(-r_t^*\right)\exp\left(-\lambda_t^{*\prime}\varepsilon_{t+1} - \frac{\lambda_t^{*\prime}\lambda_t^{*\prime}}{2}\right),$$

[46] Because the random variable $\frac{\xi_{t+1}^S}{\xi_t^S}$ is log normal with conditional mean unity, it is a likelihood ratio.

where r_t^* is now the short rate and λ_t^* is the agent's vector of risk prices. Consequently Piazzesi, Salomao, and Schneider's subjective pricing equation (15.13.3) can be expressed as

$$E^P \exp(-r_t^*) \exp\left(-\lambda_t^{*\prime}\varepsilon_{t+1} - \frac{\lambda_t^{*\prime}\lambda_t^*}{2}\right) \exp\left(-\kappa_t'\varepsilon_{t+1} - \frac{\kappa_t'\kappa_t}{2}\right) R_{j,t+1} = 1$$

or

$$E^P \exp(-r_t) \exp\left(-\left(\lambda_t^* + \kappa_t\right)'\varepsilon_{t+1} - \frac{\left(\lambda_t^* + \kappa_t\right)'\left(\lambda_t^* + \kappa_t\right)}{2}\right) R_{j,t+1} = 1 \tag{15.13.4}$$

where $r_t = r_t^* - \lambda_t^{*\prime}\kappa_t$.[47] If we compare equation (15.13.4) with the rational expectations econometrician's

$$E^P \exp(-r_t) \exp\left(-\lambda_t'\varepsilon_{t+1} - \frac{\lambda_t'\lambda_t}{2}\right) R_{j,t+1} = 1,$$

we see that what the econometrician interprets as λ_t is actually $\lambda_t^* + \kappa_t$.

Piazzesi, Salomao, and Schneider used their structure to reinterpret econometric estimates of risk prices λ_t obtained by imposing rational expectations, i.e., by mistakenly imputing the physical measure to the agents that determine asset prices via $E_t^P m_{t+1} R_{j,t+1} = 1$. When the physical measure differs from the subjective measure of the investors who price assets, investors charge risk prices λ_t^* rather than the λ_t that would be estimated by the econometrician who imposes rational expectations. Because the rational expectations econometrician's estimates of λ_t equal $\lambda_t^* + \kappa_t$, they partly reflect systematic distortions in subjective beliefs, not the representative agent's risk prices.[48]

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[47] The adjustment to the risk-free short rate r_t comes because the product of two correlated likelihood ratios is not a likelihood ratio. Such an adjustment is responsible for the difference in formulas (15.6.9) and (15.7.22) for risk-free rates of interest in the time-separable CRRA model and the Tallarini model, respectively.

[48] A closely related analysis appears in section 15.8 where we reinterpret Tallarini's (2000) model in terms of an *ex post* belief distortion that emerges from a representative agent's concerns about misspecification of the physical measure. Thus, Piazzesi, Salomao, and Schneider's analysis can be viewed as following the recommendation of Robert E. Lucas, Jr, to "look beyond high estimates of risk aversion" in order to understand high (mismeasured) prices of risk.

15.14. Concluding remarks

In this chapter, we have gone beyond chapter 9 in studying how, in the spirit of Hansen and Singleton (1983), consumer optimization alone puts restrictions on asset returns and consumption, without requiring complete markets or a fully articulated general equilibrium model. At various points in this chapter, especially in section 15.10, we have alluded to incomplete markets models. In chapters 19 and 22, we describe the ingredients of such models.

A. Riesz representation theorem

The version of the Riesz representation theorem used in this chapter is yet another ramification of population least squares regression.[49]

Let X be a Hilbert space (i.e., a complete inner product space) of random variables with inner product $< x_1, x_2 >= E(x_1 x_2)$. Let $\phi : X \to I\!R$ be a continuous linear functional mapping X into $I\!R$, the real line. We say that ϕ is *linear* because (i) for $x_1 \in X, x_2 \in X$, $\phi(x_1 + x_2) = \phi(x_1) + \phi(x_2)$, and (ii) for $a \in I\!R, x \in X, \phi(ax) = a\phi(x)$. We want to prove

Theorem 15.A.1. *(Riesz representation) Let ϕ be a continuous linear functional $\phi : X \to I\!R$. There exists a unique element $y \in X$ such that*

$$\phi(x) = E(yx). \tag{15.A.1}$$

Proof. The null space $N \equiv N(\phi)$ is defined as:

$$N(\phi) = \{x \in X : \phi(x) = 0\}.$$

$N(\phi)$ is a closed linear subspace of X. If $N = X$, then evidently $\phi(x) = 0$ for all $x \in X$. If $X \neq N(\phi)$, then there exists a non-zero vector $x_1 \in N^{\perp}$, where $N^{\perp}$ is the orthogonal complement of N, i.e., the set of vectors $y \in X$ for which $< x, y >= 0$ for every $x \in N$. In fact, $N^{\perp}$ consists of scalar multiples of one vector x_1 that is a basis for $N^{\perp}$. To prove this, assume to the contrary that there are two linearly independent vector x_1 and x_2, both of which are elements of $N^{\perp}$. The linear independence of the vectors x_1 and x_2 implies that we can

[49] See appendix D of chapter 2 for basic mechanics.